卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-EC-0202 30-B003701S-D0202

第 1 页 共 1 页
PAGE 1 OF 1

图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

电气 专业 第 02 卷 第 02 册
Electrical SPECIALITY VOLUME 02 BOOK 02

卷册名称 极1 400V站用电系统及设备安装
Volume name 400V Station Power System and Equipment Installation-Pole 1 System

批 准 APPROVED BY	2019.05.17
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	雷洋 2019.05.17
图 纸 8 张 DRAWING 8 NOS.	打印件 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 Descriptions
1	30-B003701S-D0202-001	卷册设计说明及材料表 Electrical design instructions and material table	A	1		
2	30-B003701S-D0202-002	站用电系统接线示意图 Single Line Diagram of 400V Station Power System	A	1		
3	30-B003701S-D0202-003	极1 400V站用电系统网络图 Network drawing of Pole 1 400V Station Power System	A	1		
4	30-B003701S-D0202-004	极1 400V站用电系统P1A段配置图 Configuration drawing of P1A section of 400V Station Power System	A	1		
5	30-B003701S-D0202-005	极1 400V站用电系统P1B及P1C段配置图 Configuration drawing of P1B and P1C sections of 400V Station Power System	A	1		
6	30-B003701S-D0202-006	极1 400V站用电室布置图 Layout Drawing of Pole 1 400V Station Service House	A	1		
7	30-B003701S-D0202-007	主控楼MCC1及MCC2交流分屏配置及布置图 Configuration and Layout Drawing MCC1 and MCC2 of Control Building	A	1		
8	30-B003701S-D0202-008	#1继电器室MCC1交流分屏配置及布置图 Configuration and Layout Drawing MCC1 of 500kV No.1 Relay Room	A	1		
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注:

说明：

1. 本卷册设计范围：极1 400V站用电系统、主控楼交流分屏、#1 500kV继电器室交流分屏；
2. 本卷册与其余卷册接口：（1）11kV站用电系统详见D0201卷册，11kV干式变进线电缆开列在D0201卷册。（2）公用400V站用电系统详见D0204卷册；（3）极2 400V站用电系统详见D0203卷册；（4）本卷册的动力电缆清册详见D0414卷册；（5）本卷册负责站用变屏至第1个户外检修箱之间的电缆设计，检修箱之间电缆详见D0405卷册；（6）本卷册负责站用变屏至户外照明箱之间的电缆设计，照明箱至各个灯具的电缆详见户外照明相关设计卷册；（7）本卷册负责站用变屏至二次交流电源箱之间的电缆开列，交流电源箱及其后回路设计详见二次专业相关卷册；（8）本卷册负责站用变屏至水工、消防专业相关设备动力柜（箱）的电缆设计，动力柜（箱）及其后回路设计详见相关厂家图纸；（9）本卷册负责站用电屏（或分屏）至建筑物配电箱之间的电缆设计，建筑物配电箱以下的电缆见各建筑电气卷册。
3. 400V站用电系统采用TN-C-S系统，N线在400V交流屏中直接接地。
4. 本卷册中所开列的动力电缆均为阻燃型铠装电缆，电缆敷设需满足D0406~D0410卷册的要求。
5.根据招标结果及厂家提供样本，本站交流400V采用ABB断路器，断路器型号说明如右。

E2.2 S 1250 R1250

- 框架断路器 type of switch
- 分断能力 breaking capability
  - B: 42 kA H: 100 kA
  - C: 50 kA V: 150 kA
  - N: 66 kA X: 200 kA
  - S: 85 kA
- 脱扣器额定电流 rated current of tripping device
- 断路器框架电流 rated current of switch

XT2 S 160 R160

- 塑壳断路器 type of switch
- 分断能力 breaking capability
  - B: 16 kA H: 70 kA
  - C: 25 kA L: 85 kA
  - N: 36 kA
  - S: 50 kA
- 脱扣器额定电流 rated current of tripping device
- 断路器框架电流 rated current of switch

Remarks:

1.The design scope of this volume includes: 400V station power system-Pole 1 system;Motor control center(MCC) of control building;MCC of operation and maintenance area;MCC of 500kV No.1 Relay Room;
2.The interface between this volume and other volumes:(1)The incoming cables of 11kV transformer are listed in D0201;(2)400V station power system -Utility System refers to D0204;(3)400V station power system -Pole 2 System refers to D0203;(4)All power cables used in this volume are listed in D0414;(5)The cable from switch cabinet to first outdoor maintenance box is designed by this volume,while cables between maintenance boxes are designed by D0405;(6)The cable from switch cabinet to outdoor lighting box is designed by this volume,while cables from outdoor lighting box to lamps and lanterns are designed by D0401 and D0402 respectively;(7)The cables from switch cabinet to secondary electrical boxes are designed by this volume,while secondary electrical boxes are designed by secondary electrical major;(8)The cables from switch cabinet to power boxes of irrigation works are designed by this volume,while power boxes are designed by manufacture;(9)The cables from switch cabinet to distribution boxes of buildings are designed by this volume,while distribution boxes are designed by other volumes respectively.
3.TN-C-S system is adopt by 400V station power system.
4.All cables designed by this volume are inflaming retarding and armoured,cable laying should according to D0406~D0410.
5.ABB switch are adopted。

材料汇总表
Material table

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	备注 Comments
1	11kV干式配电变压器 11kV Dry type Transformer	SCZB11-2500/11，11±4x2.5%/0.4kV 2500kVA;Uk=8%;Dyn11	台 set		开列在D0204卷册 Refer to D0204
2	400V交流开关柜 400V switch cabinet	抽屉式，Drawer type	面 set	18	[深圳市泰昂能源科技股份有限公司] [Shenzhen Tieon]
3	400V交流分屏 400V MCC	固定式，Fixed type	面 set	3	[深圳市泰昂能源科技股份有限公司] [Shenzhen Tieon]
4	母线桥 Bus bridge	4000A	米 m	5	[深圳市泰昂能源科技股份有限公司] [Shenzhen Tieon]
5	母线桥 Bus bridge	1250A	米 m	5	[深圳市泰昂能源科技股份有限公司] [Shenzhen Tieon]
6	铜绞线 copper stranded conductor	150mm²	米 m	210	
7	镀锡铜鼻子 copper cable lug	DTM 150	套 set	14	
8	放热焊点 welding point	TJ150与TJ120之间 for welding of TJ150 and TJ120	套 set	14	
9	动力电缆 power cable	ZB-YJV22-0.6/1.0kV ZB-YJVP22-0.6/1.0kV	米 m		开列在D0414卷册 Refer to D0414

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

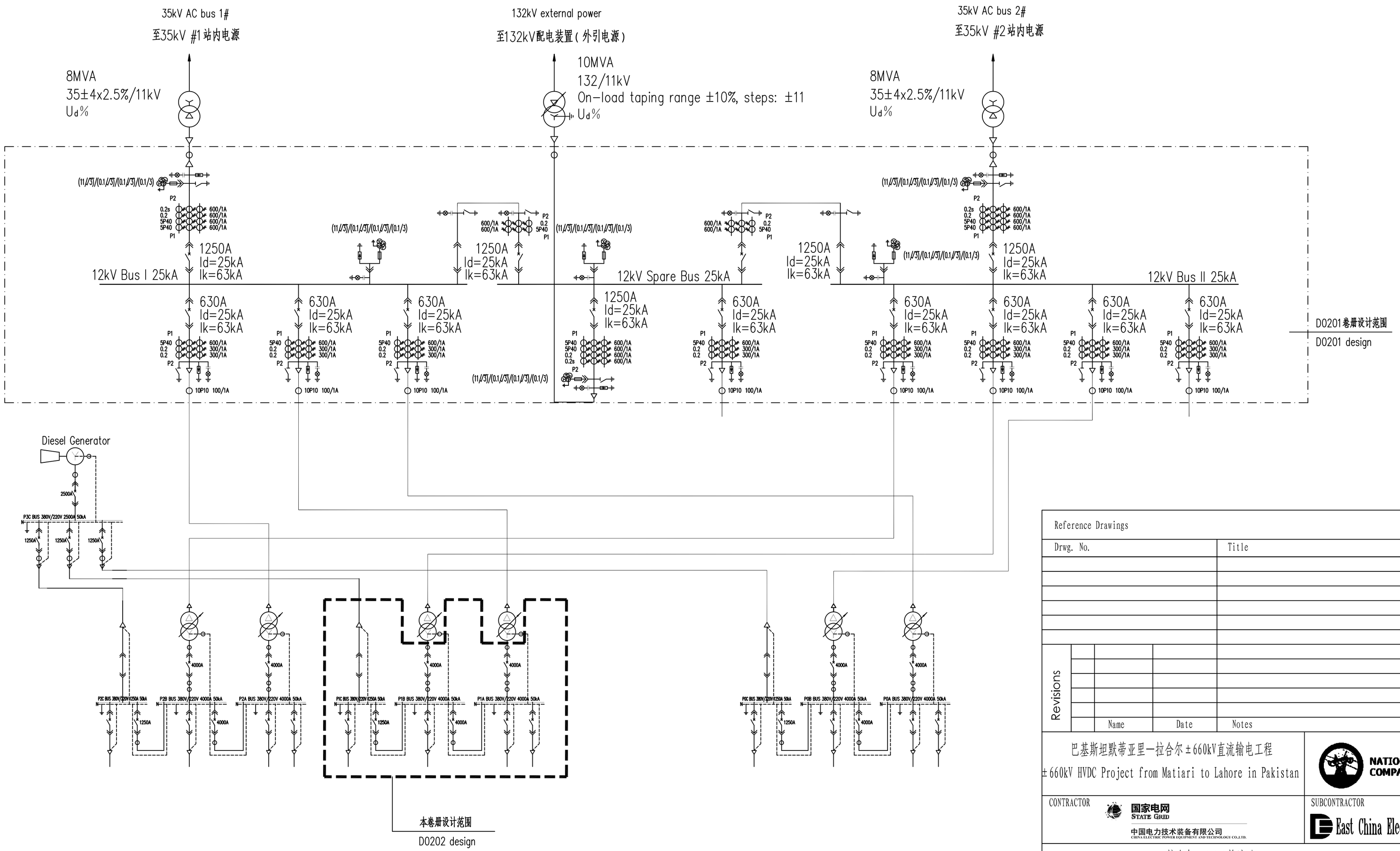
M L H V D C - L C S - E C I - D D - E C - 0 2 0 2 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	李建辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益平	2019.05.17

卷册设计说明及材料表
Electrical design instructions and material table

比例 Scale: | 图号 DWG NO.: 30-B003701S-D0202-001 | 版本 REV.: A

297X420



Reference Drawings

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Revisions

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巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 2 0 2 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	李建辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益竿	2019.05.17

站用电系统接线示意图

Single Line Diagram of Station Power System

比例 Scale:

图号 DWG NO.: 30-B003701S-D0202-002

版本 REV.: A

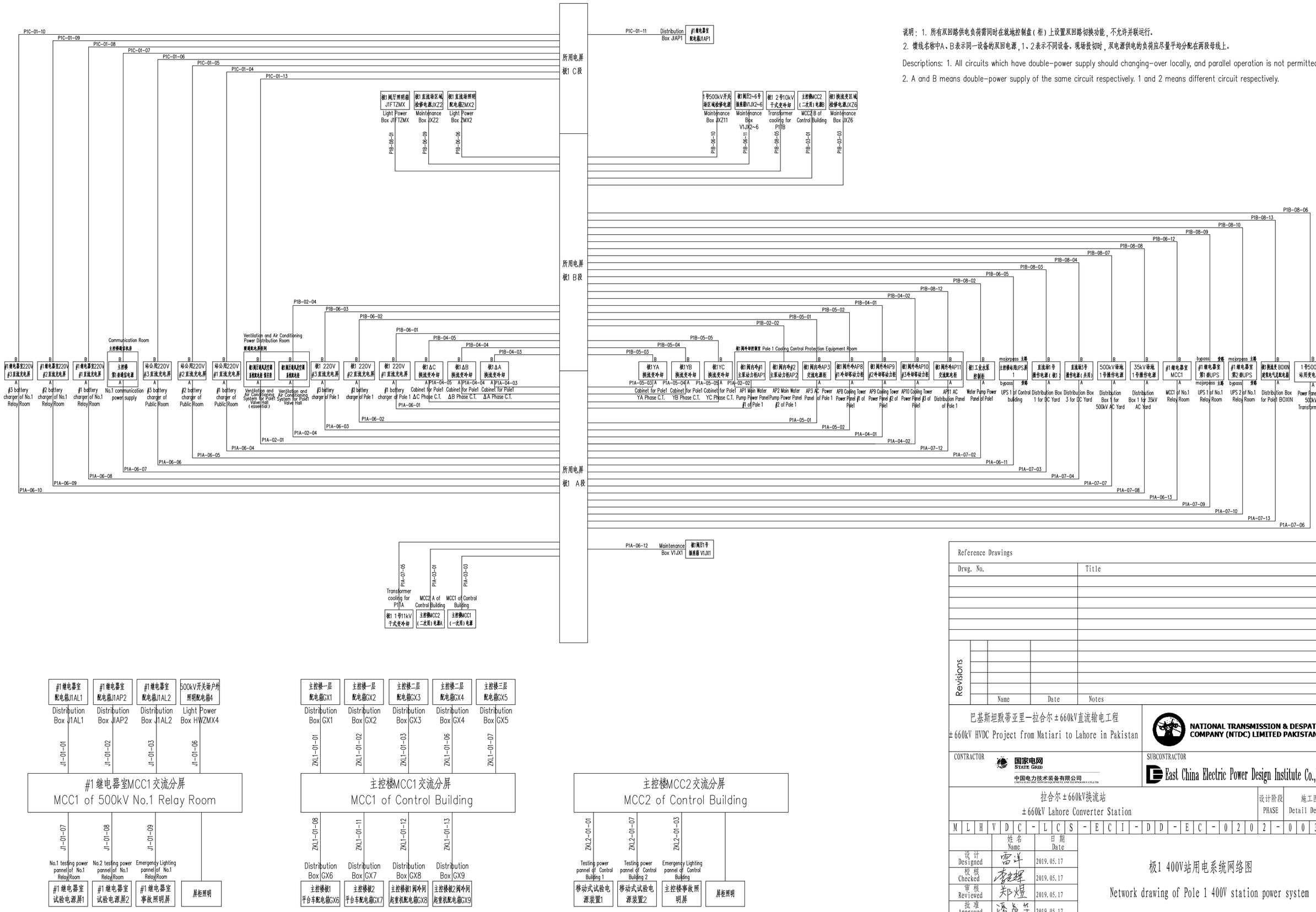
420X594

说明：1. 所有双回路供电负荷需同时在就地控制盘（柜）上设置双回路切换功能，不允许并联运行。

2. 馈线名称中A、B表示同一设备的双回电源，1、2表示不同设备。现场投切时，双电源供电的负荷应尽量平均分配在两段母线上。

Descriptions: 1. All circuits which have double-power supply should changing-over locally, and parallel operation is not permitted.

2. A and B means double-power supply of the same circuit respectively. 1 and 2 means different circuit respectively.



Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 2 0 2 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	李建辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益竹	2019.05.17

极1 400V站用电系统网络图

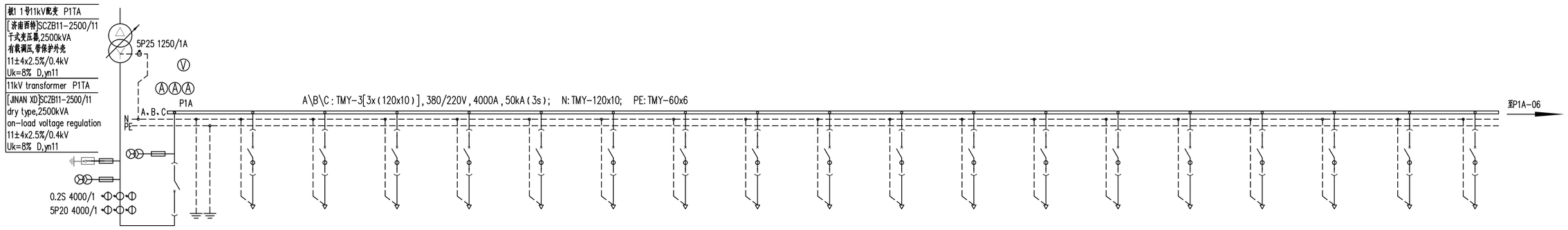
Network drawing of Pole 1 400V station power system

比例 Scale:

图号 DWG NO.: 30-B003701S-D0202-003

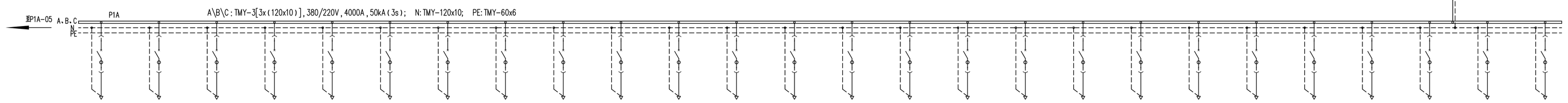
版本 REV.: A

594X841



配电屏编号	P1A-01	P1A-02				P1A-03				P1A-04					P1A-05				
屏宽(mm)	1000	650				650				650					650				
回路名称	极1 1号11kV配变进线 No.1 Pole 1 11kV transformer incoming line	极1洞厅通风及空调系统配电柜 保安段 A Ventilation and Air Conditioning System for Pole1 Valve Hall (Essential) A	极1强排水 #1主泵驱动柜AP1 AP1 Main Water Pump Power Panel #1 of Pole 1	备用 for spare	极1洞厅通风及空调系统配电柜 A Ventilation and Air Conditioning System for Pole1 Valve Hall A	主控楼MCC2 (二次用)电源A MCC2 of Control Building A	备用 for spare	主控楼MCC1 (一次用)电源 MCC1 of Control Building	备用 for spare	极1洞外冷AP9 #2冷却塔动力柜 A AP9 Cooling Tower Power Panel #2 of Pole1 A	极1洞外冷AP10 #3冷却塔动力柜 A AP10 Cooling Tower Power Panel #3 of Pole1 A	极1 ΔA换流变冷却 A Cabinet for Pole1 ΔA Phase C.T. A	极1 ΔB换流变冷却 A Cabinet for Pole1 ΔB Phase C.T. A	极1 ΔC换流变冷却 A Cabinet for Pole1 ΔC Phase C.T. A	极1洞内冷AP3 交流盘屏柜 A AP3 AC Power Panel of Pole1 A	极1洞外冷AP8 #1冷却塔动力柜 A AP8 Cooling Tower Power Panel #1 of Pole1 A	极1 YA换流变冷却 A Cabinet for Pole1 YA Phase C.T. A	极1 YB换流变冷却 A Cabinet for Pole1 YB Phase C.T. A	极1 YC换流变冷却 A Cabinet for Pole1 YC Phase C.T. A
回路额定容量(kW)		50	110		408.2					60	60	105	105	105	70	60	105	105	105
回路额定工作电流(A)	3798	108.2	205		734.18	~120		~400		120	120	270	270	270	140	120	270	270	270
断路器型号	E4.2S4000 R4000 3P	XT2S160 R160	T5S400 R400	T5S630 R630	E2.2S1250 R1250	XT4S250 R250	T5S400 R400	T5S630 R630	E2.2S1250 R1250	XT2S160 R160	XT2S160 R160	T5S400 R400	T5S400 R400	T5S400 R400	XT4S250 R250	XT2S160 R160	T5S400 R400	T5S400 R400	T5S400 R400
断路器额定电流Iu(A)	4000	160	400	630	1250	250	400	630	1250	160	160	400	400	400	250	160	400	400	400
保护器型号	T-LSIG	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	T-LSI	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	T-LSI	LS/I	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	PR222DS/P-LSI	LS/I	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	PR222DS/P-LSI
保护器额定电流In(A)	4000	160	400		1250	250		630		160	160	400	400	400	250	160	400	400	400
保护电器 瞬时或短延时脱扣电流值I1/脱扣曲线长延时特性(A/s)	3900/3	120/3	240/3		875/3	180/3		504/3		144/3	144/3	320/3	320/3	320/3	160/3	144/3	320/3	320/3	320/3
保护电器 瞬时或短延时脱扣电流值I2/脱扣曲线短延时特性(A/s)	12000/0.6(t=k)	400/0.1	4000/0.1		5000/0.1	1000/0.1		3150/0.1		1600/0.1	1600/0.1	1600/0.1	1600/0.1	1600/0.1	2000/0.1	1600/0.1	1600/0.1	1600/0.1	1600/0.1
保护电器 建议瞬动整定电流I3(A)	关闭 Closed	关闭 Closed	关闭 Closed		关闭 Closed	关闭 Closed		关闭 Closed		关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed
保护电器 接地保护瞬时或长延时脱扣电流值I4/脱扣曲线长延时特性(A/s)	1200/0.8																		
电流互感器变比	4000/1	150/1	400/1		1200/1	250/1		600/1		150/1	150/1	400/1	400/1	400/1	250/1	150/1	400/1	400/1	400/1
电缆型号及芯数x截面(mm²)		ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3X185+1X95		ZRB-YJV22-0.6/1.0 3//(3X240+1X120)	ZRB-YJV22-0.6/1.0 -3x95+1x50		ZRB-YJV22-0.6/1.0 2//(3X185+1X95)		ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120
电缆载流量(空气中50℃)		181.6	276		974.4	181.6		552		181.6	181.6	324.8	324.8	324.8	181.6	181.6	324.8	324.8	324.8
电缆编号(回路编号)		P1A-02-01	P1A-02-02	P1A-02-03	P1A-02-04	P1A-03-01	P1A-03-02	P1A-03-03	P1A-03-04	P1A-04-01	P1A-04-02	P1A-04-03	P1A-04-04	P1A-04-05	P1A-05-01	P1A-05-02	P1A-05-03	P1A-05-04	P1A-05-05
电缆长度(米)		110	35		110x3	39		38x2		39	40	92	80	69	35	42	107	120	132

分段引线,母线桥上进线
引自P1B-07屏母线



配电屏编号	P1A-06															P1A-07												
屏宽(mm)	650															1000												
回路名称	极1 220V #1直流充电屏 A #1 battery charger of Pole 1 A	极1 220V #2直流充电屏 A #2 battery charger of Pole 1 A	极1 220V #3直流充电屏 A #3 battery charger of Pole 1 A	站公用220V #1直流充电屏 A #1 battery charger of Public Room A	站公用220V #2直流充电屏 A #2 battery charger of Public Room A	站公用220V #3直流充电屏 A #3 battery charger of Public Room A	主控楼 第1套通信电源 A No.1 communication power supply A	#1继电器室220V #1直流充电屏 A #1 battery charger of No.1 Relay Room A	#1继电器室220V #2直流充电屏 A #2 battery charger of No.1 Relay Room A	#1继电器室220V #3直流充电屏 A #3 battery charger of No.1 Relay Room A	主控楼 站用UPS第1步幕 UPS 1 of Control building bypass	极1洞口号 检修箱 V1JX1 Maintenance Box V1JX1	#1继电器室MCC1 A MCC1 of No.1 Relay Room A	备用 for spare	极1工业水泵控制柜 A Water Pump Power Panel of Pole1 A	直流屏1号 #1 A Distribution Box 1 for DC Yard A	直流屏3号 #3 A Distribution Box 3 for DC Yard A	极1 1号11kV干式变冷却 Transformer cooling for P1TA	1号500kV 站用变电源盘 A Power Panel for 500kV Transformer 1 A	500kV场地 1号接线电源 A Distribution Box 1 for 500kV AC Yard A	35kV场地 1号接线电源 A Distribution Box 1 for 35kV AC Yard A	#1继电器室 第1套UPS 主路 UPS 1 of No.1 Relay Room majorpass	#1继电器室 第2套UPS 旁路 UPS 2 of No.1 Relay Room bypass	备用 for spare	极1洞外冷AP11 交流配电盘 A AP11 AC Distribution Panel of Pole 1 A	极1换流变 BOXIN 就地电气总配电盘 A Distribution Box for Pole1 BOXIN A
回路额定容量(kW)	26.4	26.4	26.4	44	44	44	20	44	44	44	30(单相)					11	4	4	4	3	3	3	3			45	50
回路额定工作电流(A)	~47	~47	~47	~79	~79	~79	~35	~79	~79	~79	~145	~120	~180			22	80	80	~12	32	63	63	18	18		90	~100
断路器型号	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT4S250 R250	XT2S160 R160	XT4S250 R250		XT2S160 R100	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R25	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R63	XT2S160 R63	XT2S160 R160	XT2S160 R160	XT2S160 R160
断路器额定电流Iu(A)	160	160	160	160	160	160	160	160	160	160	250	160	250		160	160	160	160	160	160	160	160	160	160	160	160	160
保护器型号	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I		LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I
保护器额定电流In(A)	100	100	100	100	100	100	63	100	100	100	250	160	250			63	100	100	25	63	100	100	63	63		160	160
保护电器 瞬时或短延时脱扣电流值I1/脱扣曲线长延时特性(A/s)	55/3	55/3	55/3	90/3	90/3	90/3	44.1/3	90/3	90/3	90/3	160/3	144/3	200/3			63/3	90/3	90/3	20/3	63/3	75/3	75/3	25.2/3	25.2/3		112/3	120/3
保护电器 瞬时或短延时脱扣电流值I2/脱扣曲线短延时特性(A/s)	500/0.1	500/0.1	500/0.1	300/0.1	300/0.1	300/0.1	189/0.1	300/0.1	300/0.1	300/0.1	1000/0.1	800/0.1	500/0.1			189/0.1	250/0.1	250/0.1	关闭 Closed	189/0.1	200/0.1	200/0.1	126/0.1	126/0.1		1600/0.1	480/0.1
保护电器 建议瞬动整定电流I3(A)	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed			关闭 Closed	关闭 Closed	关闭 Closed	250	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed		关闭 Closed	关闭 Closed
保护电器 接地保护瞬时或长延时脱扣电流值I4/脱扣曲线长延时特性(A/s)																											
电流互感器变比	100/1	100/1	100/1	100/1	100/1	100/1	75/1	100/1	100/1	100/1	250/1	150/1	250/1			75/1	100/1	100/1	30/1	75/1	100/1	100/1	75/1	75/1		150/1	150/1
电缆型号及芯数x截面(mm²)	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -2x95(缺A相)	ZRB-YJVP22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x120+1x70		ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -2x10(缺B相)	ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -2x10(缺B相)	ZRB-YJV22-0.6/1.0 -2x10(缺C相)		ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x50+1x25
电缆载流量(空气中50℃)	63.2	63.2	63.2	100	100	100	63.2	121.6	121.6	121.6	188.8	181.6	210.4		83.2	121.6	121.6	53.6	83.2	100	100	53.6	53.6			121.6	121.6
电缆编号(回路编号)	P1A-06-01	P1A-06-02	P1A-06-03	P1A-06-04	P1A-06-05	P1A-06-06	P1A-06-07	P1A-06-08	P1A-06-09	P1A-06-10	P1A-06-11	P1A-06-12	P1A-06-13	P1A-07-01	P1A-07-02	P1A-07-03	P1A-07-04	P1A-07-05	P1A-07-06	P1A-07-07	P1A-07-08	P1A-07-09	P1A-07-10	P1A-07-11	P1A-07-12	P1A-07-13	
电缆长度(米)	37	34	38	66	73	67	69	162	155	163	38	27	159		245	212	135	19	196	271	179	158	159		40	115	
	注9 Note9	注9 Note9	注9 Note9															注9 Note9									

说明:1. 两台干式配电变压器不得并列运行。
2. 站用电屏本期一次配置完成,采用抽屉式。
3. 所有馈线均应装设带电显示器,进线及分段回路装设电压表、电流表、功率表及带电显示器。
4. 有关站用电的控制、保护、信号及联闭锁要求见二次专业相关卷册。
5. 除特别注明"t=k"外,本表中断路器所选取的脱扣曲线均为反时限。
6. PR222长延时脱扣曲线时间档位所对应的电流值为@6I1,短延时脱扣曲线时间档位所对应的电流值为@8In。
7. LS/I长延时脱扣曲线时间档位所对应的电流值为@3I1,短延时脱扣曲线时间档位所对应的电流值为@10In。
8. T-LSI/T-LSIG长延时脱扣曲线时间档位所对应的电流值为@3I1,短延时脱扣曲线时间档位所对应的电流值为@10In,接地保护脱扣曲线时间档位所对应的电流值为@4In。
9. 为满足电缆热稳定要求,标注处电缆敷设长度需按图示要求,可考虑迂回敷设等方式。
10. 电流载流量根据重庆泰山电缆有限公司提供的资料确定(按空气中50°环温考虑,并考虑0.8的敷设系数)。
11. 图中开关保护整定值仅供参考,现场可根据实际情况调整。

Descriptions:1. Parallel operation of two 11kV distribution transformer is not permitted.
2. Circuit breakers in all switch cabinets are drawer type.
3. The meters should be configured accordingly.
4. The design of control and relay protection refer to secondary electrical major.
5. All tripping curve of circuit breakers are inverse time-lag type except for "t=k, which means constant time-lag.
6. The time of long delay overload tripping curve of PR222 is @6I1, the time of short delay overload tripping curve of PR222 is @8In.
7. The time of long delay overload tripping curve of LS/I is @3I1, the time of short delay overload tripping curve of LS/I is @10In.
8. The time of long delay overload tripping curve of T-LSI/T-LSIG is @3I1, the time of short delay overload tripping curve of T-LSI/T-LSIG is @10In, the time of ground protection tripping curve of T-LSIG is @4In.
9. The length of cables should be laid according to the drawing.
10. Current-carrying capacity of cable is according to cable manufacture.
11. The settings of tripping device are for reference.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

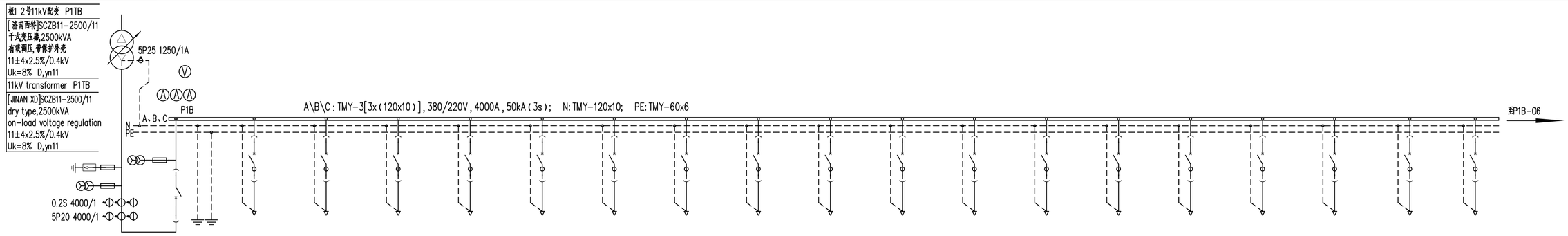
M L H V D C - L C S - E C I - D D - E C - 0 2 0 2 - 0 0 4 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	凌志辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益竹	2019.05.17

极1 400V站用电系统P1A段配置图
Configuration drawing of P1A section of 400V Station Power System

比例 Scale | 图号 DWG NO. 30-B003701S-D0202-004 | 版本 REV. A

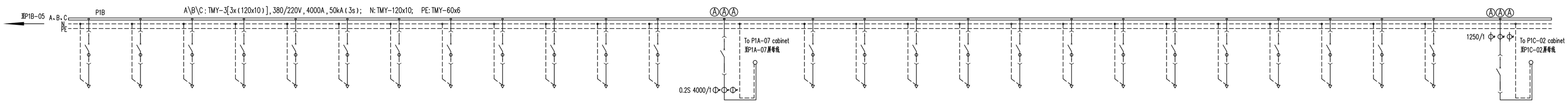
594X841



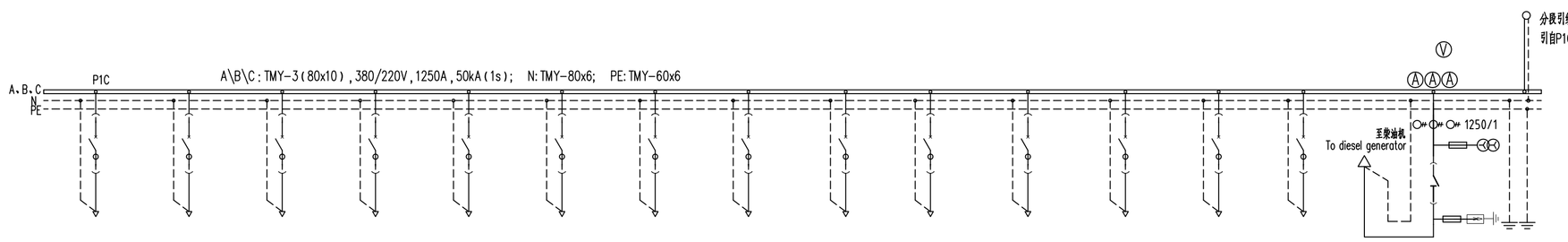
配电屏编号	P1B-01	P1B-02				P1B-03				P1B-04					P1B-05				
屏宽(mm)	1000	650				650				650					650				
回路名称	极1 2号11kV取发进线 No.2 Pole 1 11kV transformer incoming line	备用 for spare	极1空调冷冻 极2主蒸汽冷凝AP2 AP2 Main Water Pump Power Panel of Pole 1	备用 for spare	极1洞厅通风及空调系统配电柜 B Ventilation and Air Conditioning System for Pole1 Valve Hall B	主控楼MCC2 (二次用)电源B MCC2 of Control Building B	备用 for spare	极1换流变区域检修电源 JXZ6 Maintenance Box JXZ6	备用 for spare	极1洞外冷AP9 极2冷却塔动力柜 B AP9 Cooling Tower Power Panel of Pole1 B	极1洞外冷AP10 极3冷却塔动力柜 B AP10 Cooling Tower Power Panel of Pole1 B	极1 ΔA换流变冷却 B Cabinet for Pole1 ΔA Phase C.T. B	极1 ΔB换流变冷却 B Cabinet for Pole1 ΔB Phase C.T. B	极1 ΔC换流变冷却 B Cabinet for Pole1 ΔC Phase C.T. B	极1洞外冷AP3 交流电源柜 B AP3 AC Power Panel of Pole 1 B	极1洞外冷AP8 极1冷却塔动力柜 B AP8 Cooling Tower Power Panel of Pole1 B	极1 YA换流变冷却 B Cabinet for Pole1 YA Phase C.T. B	极1 YB换流变冷却 B Cabinet for Pole1 YB Phase C.T. B	极1 YC换流变冷却 B Cabinet for Pole1 YC Phase C.T. B
回路额定容量(kW)			110		408.2					60	60	105	105	105	70	60	105	105	105
回路额定工作电流(A)	3798		205		734.18	~120		~500		120	120	270	270	270	140	120	270	270	270
断路器型号	E4.2S4000 R4000 3P	XT4S250 R250	T5S400 R400	T5S630 R630	E2.2S1250 R1250	XT4S250 R250	T5S400 R400	T5S630 R630	E2.2S1250 R1250	XT2S160 R160	XT2S160 R160	T5S400 R400	T5S400 R400	T5S400 R400	XT4S250 R250	XT2S160 R160	T5S400 R400	T5S400 R400	T5S400 R400
断路器额定电流 Iu(A)	4000	250	400	630	1250	250	400	630	1250	160	160	400	400	400	250	160	400	400	400
保护电器 脱扣器型号	T-LSIG	LS/I	PR222DS/P-LSI	PR2220S/P-LSI	T-LSI	LS/I	PR2220S/P-LSI	PR222DS/P-LSI	T-LSI	LS/I	LS/I	PR2220S/P-LSI	PR222DS/P-LSI	PR222DS/P-LSI	LS/I	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	PR2220S/P-LSI
保护电器 脱扣器额定电流 In(A)	4000		400		1250	250		630		160	160	400	400	400	250	160	400	400	400
保护电器 建议长延时脱扣门限值 I1/脱扣曲线时间挡位(A/s)	3900/3		240/3		875/3	180/3		630/3		144/3	144/3	320/3	320/3	320/3	160/3	144/3	320/3	320/3	320/3
保护电器 建议短延时脱扣门限值 I2/脱扣曲线时间挡位(A/s)	12000/0.6(t=k)		4000/0.1		5000/0.1	1000/0.1		1890/0.1		1600/0.1	1600/0.1	1600/0.1	1600/0.1	1600/0.1	2000/0.1	1600/0.1	1600/0.1	1600/0.1	1600/0.1
保护电器 建议瞬动整定电流 I3(A)	关闭 Closed		关闭 Closed		关闭 Closed	关闭 Closed		关闭 Closed		关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed
保护电器 建议接地保护门限值 I4/脱扣曲线时间挡位(A/s)	1200/0.8																		
电流互感器变比	4000/1		400/1		1200/1	250/1		600/1		150/1	150/1	400/1	400/1	400/1	250/1	150/1	400/1	400/1	400/1
电缆型号及芯数x截面(mm²)			ZRB-YJV22-0.6/1.0 -3X185+1X95		ZRB-YJV22-0.6/1.0 3//(3X240+1X120)	ZRB-YJV22-0.6/1.0 -3x95+1x50		ZRB-YJV22-0.6/1.0 2//(3X240+1X120)		ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120
电缆载流量(空气中50℃)			276		974.4	181.6		649.6		181.6	181.6	324.8	324.8	324.8	181.6	181.6	324.8	324.8	324.8
电缆编号(回路编号)		P1B-02-01	P1B-02-02	P1B-02-03	P1B-02-04	P1B-03-01	P1B-03-02	P1B-03-03	P1B-03-04	P1B-04-01	P1B-04-02	P1B-04-03	P1B-04-04	P1B-04-05	P1B-05-01	P1B-05-02	P1B-05-03	P1B-05-04	P1B-05-05
电缆长度(米)			28		104x3	39		115x2		30	31	99	88	76	27	30	114	128	139

说明:1. 两台干式配电变压器不得并列运行。
2. 站用电屏本期一次配置完成,采用抽屉式。
3. 所有馈线均应装设带电显示器,进线及分段回路装设电压表、电流表、功率表及带电显示器。
4. 有关站用电的控制、保护、信号及联闭锁要求见二次专业相关卷册。
5. 除特别注明"t=k"外,本表中断路器所选取的脱扣曲线均为反时限。
6. PR222长延时脱扣曲线时间档位所对应的电流值为@6I1,短延时脱扣曲线时间档位所对应的电流值为@8In。
7. LS/I长延时脱扣曲线时间档位所对应的电流值为@3I1,短延时脱扣曲线时间档位所对应的电流值为@10In。
8. T-LSI/T-LSIG长延时脱扣曲线时间档位所对应的电流值为@3I1,短延时脱扣曲线时间档位所对应的电流值为@10In,接地保护脱扣曲线时间档位所对应的电流值为@4In。
9. 为满足电缆热稳定要求,标注处电缆敷设长度需按图示要求,可考虑迂回敷设等方式。
10. 电流载流量根据重庆泰山电缆有限公司提供的资料确定(按空气中50°环温考虑,并考虑0.8的敷设系数)。
11. 图中开关保护整定值仅供参考,现场可根据实际情况调整。

Descriptions:1. Parallel operation of two 11kV distribution transformer is not permitted.
2. Circuit breakers in all switch cabinets are drawer type.
3. The meters should be configured accordingly.
4. The design of control and relay protection refer to secondary electrical major.
5. All tripping curve of circuit breakers are inverse time-lag type except for "t=k, which means constant time-lag.
6. The time of long delay overload tripping curve of PR222 is @6I1, the time of short delay overload tripping curve of PR222 is @8In.
7. The time of long delay overload tripping curve of LS/I is @3I1, the time of short delay overload tripping curve of LS/I is @10In.
8. The time of long delay overload tripping curve of T-LSI/T-LSIG is @3I1, the time of short delay overload tripping curve of T-LSI/T-LSIG is @10In, the time of ground protection tripping curve of T-LSIG is @4In.
9. The length of cables should be laid according to the drawing.
10. Current-carrying capacity of cable is according to cable manufacture.
11. The settings of tripping device are for reference.



配电屏编号	P1B-06													P1B-07	P1B-08													P1C-03
屏宽(mm)	650													1000	650													1000
回路名称	极1 220V 极1直流充电屏B 极1 battery charger of Pole 1 B	极1 220V 极2直流充电屏B 极2 battery charger of Pole 1 B	极1 220V 极3直流充电屏B 极3 battery charger of Pole 1 B	备用 for spare	主控楼 站用UPS屏1 主路 UPS 1 of Control building mainpass	极1 直流场照明 配电箱 ZMX2 Light Power Box ZMX2	备用 for spare	备用 for spare	极1 直流场 区域检修电源 JXZ2 Maintenance Box JXZ2	1号500kV开关场 区域检修电源 JXZ11 Maintenance Box JXZ11	极1洞区2-6号 轴流风机JX2~6 Maintenance Box VLJX2~6	极1 电机房MCC1 B MCC1 of No.1 Relay Room B	分段 section switch	极1洞厅通风设备 JHT7ZMX Light Power Box JHT7ZMX	极1 工业水泵控制柜 B Water Pump Power Panel of Pole1 B	直流屏1号 蓄电电源(极1) B Distribution Box 1 for DC Yard B	直流屏3号 蓄电电源(共用) B Distribution Box 3 for DC Yard B	极1 2号11kV 干式变冷却 Transformer cooling for P1TB	1号500kV 站用变电源屏B Power Panel for 500kV Transformer 1 B	500kV母线 1号蓄电电源 B Distribution Box 1 for 500kV AC Yard B	35kV母线 1号蓄电电源 B Distribution Box 1 for 35kV AC Yard B	极1 电机房室 极1 单UPS 旁路 UPS 1 of No.1 Relay Room bypass	极1 电机房室 极2 单UPS 主路 UPS 2 of No.1 Relay Room mainpass	备用 spare	极1洞外冷AP11 交流配电柜 B AP11 AC Distribution Panel of Pole 1 B	极1 换流变 BOXIN 液氨电气控制电源 B Distribution Box for Pole1 BOXIN B	除油机与工作母线分段 section switch
回路额定容量(kW)	26.4	26.4	26.4		30										20.42	11				3		3			45	50		
回路额定工作电流(A)	~47	~47	~47		~54	40			~160	~160	~160	~180	3798		53.06	22	80	80	~12	32	63	63	~18	~18		90	~100	
断路器型号	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT4S250 R250	XT4S250 R250	XT4S250 R250	XT4S250 R250	E4.2S4000 R4000 3P		XT2S160 R100	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R25	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R63	XT2S160 R63	XT2S160 R100	XT2S160 R160	XT2S160 R160	E2.2S1250 R1250 3P
断路器额定电流 Iu(A)	160	160	160	160	160	160	160	160	250	250	250	250	4000		160	160	160	160	160	160	160	160	160	160	160	160	160	1250
保护电器 脱扣器型号	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	T-LSI		LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	T-LSI
保护电器 脱扣器额定电流 In(A)	100	100	100		100	100			250	250	250	250	4000		100	63	100	100	25	63	100	100	63	63		160	160	1250
保护电器 建议长延时脱扣门限值 I1/脱扣曲线时间挡位(A/s)	55/3	55/3	55/3		70/3	50/3			160/3	160/3	160/3	200/3	3900/3		60/3	63/3	90/3	90/3	20/3	63/3	75/3	75/3	25.2/3	25.2/3		112/3	120/3	1200/3
保护电器 建议短延时脱扣门限值 I2/脱扣曲线时间挡位(A/s)	500/0.1	500/0.1	500/0.1		300/0.1	200/0.1			500/0.1	500/0.1	500/0.1	500/0.1	10000/0.4(t=k)		300/0.1	189/0.1	250/0.1	250/0.1	关闭 Closed	189/0.1	200/0.1	200/0.1	126/0.1	126/0.1		1600/0.1	480/0.1	10000/0.4(t=k)
保护电器 建议瞬动整定电流 I3(A)	关闭 Closed	关闭 Closed	关闭 Closed		关闭 Closed	关闭 Closed			关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed		关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	250	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed		关闭 Closed	关闭 Closed	关闭 Closed
保护电器 建议接地保护门限值 I4/脱扣曲线时间挡位(A/s)													1200/0.8															500/0.4
电流互感器变比	100/1	100/1	100/1		100/1	100/1			250/1	250/1	250/1	250/1	4000/1		100/1	75/1	100/1	100/1	30/1	75/1	100/1	100/1	75/1	75/1		150/1	150/1	1250/1
电缆型号及芯数x截面(mm²)	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x16+1x10		ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -3x16+1x10			ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJVP22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x120+1x70	母线桥		ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -2x10(拼C相)	ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -2x10(拼A相)	ZRB-YJV22-0.6/1.0 -2x10(拼B相)		ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x50+1x25	母线桥
电缆载流量(空气中50℃)	63.2	63.2	63.2		100	63.2			181.6	324.8	181.6	210.4			83.2	83.2	121.6	121.6	53.6	83.2	100	100	53.6	53.6		121.6	121.6	
电缆编号(回路编号)	P1B-06-01	P1B-06-02	P1B-06-03	P1B-06-04	P1B-06-05	P1B-06-06	P1B-06-07	P1B-06-08	P1B-06-09	P1B-06-10	P1B-06-11	P1B-06-12			P1B-08-01	P1B-08-02	P1B-08-03	P1B-08-04	P1B-08-05	P1B-08-06	P1B-08-07	P1B-08-08	P1B-08-09	P1B-08-10	P1B-08-11	P1B-08-12	P1B-08-13	
电缆长度(米)	28	32	27		38	104			105	280	61	152			102	236	205	122	19	188	264	175	152	153		30	122	
																			备注 Note9							备注 Note9		



配电屏编号	P1C-01															P1C-02
屏宽(mm)	650															1000
回路名称	备用 for spare	备用 for spare	备用 for spare	站公用220V 极1直流充电屏B 极1 battery charger of Public Room B	站公用220V 极2直流充电屏B 极2 battery charger of Public Room B	站公用220V 极3直流充电屏B 极3 battery charger of Public Room B	主控楼 第1通信电源B No.1 communication power supply B	极1电机房室220V 极1直流充电屏B 极1 battery charger of No.1 Relay Room B	极1电机房室220V 极2直流充电屏B 极2 battery charger of No.1 Relay Room B	极1电机房室220V 极3直流充电屏B 极3 battery charger of No.1 Relay Room B	极1电机房室 配电箱JIAP1 Distribution Box JIAP1	备用 for spare	极1洞厅通风及空调系统配电柜 保安段 B Ventilation and Air Conditioning System for Pole1 Valve Hall (Essential) B	备用 for spare	柴油机进线 incoming line of diesel generator
回路额定容量(kW)				44	44	44	20	44	44	44	25		50		359
回路额定工作电流(A)				~79	~79	~79	~35	~79	~79	~79	38		108.2		655.2
断路器型号	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R160	XT4S250 R250	隔离开关 disconnector
断路器额定电流 Iu(A)	160	160	160	160	160	160	160	160	160	160	160	160	160	250	E2.2S/MS/1250/R1250/3P
保护电器 脱扣器型号	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	
保护电器 脱扣器额定电流 In(A)				100	100	100	63	100	100	100	100		160		
保护电器 建议长延时脱扣门限值 I1/脱扣曲线时间挡位(A/s)				90/3	90/3	90/3	44.1/3	90/3	90/3	90/3	50/3		120/3		
保护电器 建议短延时脱扣门限值 I2/脱扣曲线时间挡位(A/s)				300/0.1	300/0.1	300/0.1	189/0.1	300/0.1	300/0.1	300/0.1	200/0.1		400/0.1		
保护电器 建议瞬动整定电流 I3(A)				关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed		关闭 Closed		
保护电器 建议接地保护门限值 I4/脱扣曲线时间挡位(A/s)															
电流互感器变比				100/1	100/1	100/1	75/1	100/1	100/1	100/1	100/1		150/1		
电缆型号及芯数x截面(mm²)				ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x16+1x10		ZRB-YJV22-0.6/1.0 -3x95+1x50		ZRB-YJV22-0.6/1.0 3//(3X240+1X120)
电缆载流量(空气中50℃)				100	100	100	63.2	121.6	121.6	121.6	63.2		181.6		974.4
电缆编号(回路编号)	P1C-01-01	P1C-01-02	P1C-01-03	P1C-01-04	P1C-01-05	P1C-01-06	P1C-01-07	P1C-01-08	P1C-01-09	P1C-01-10	P1C-01-11	P1C-01-12	P1C-01-13	P1C-01-14	P1C-02-01
电缆长度(米)				63	71	65	67	164	158	165	145		105		233x3

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 2 0 2 - 0 0 5 A

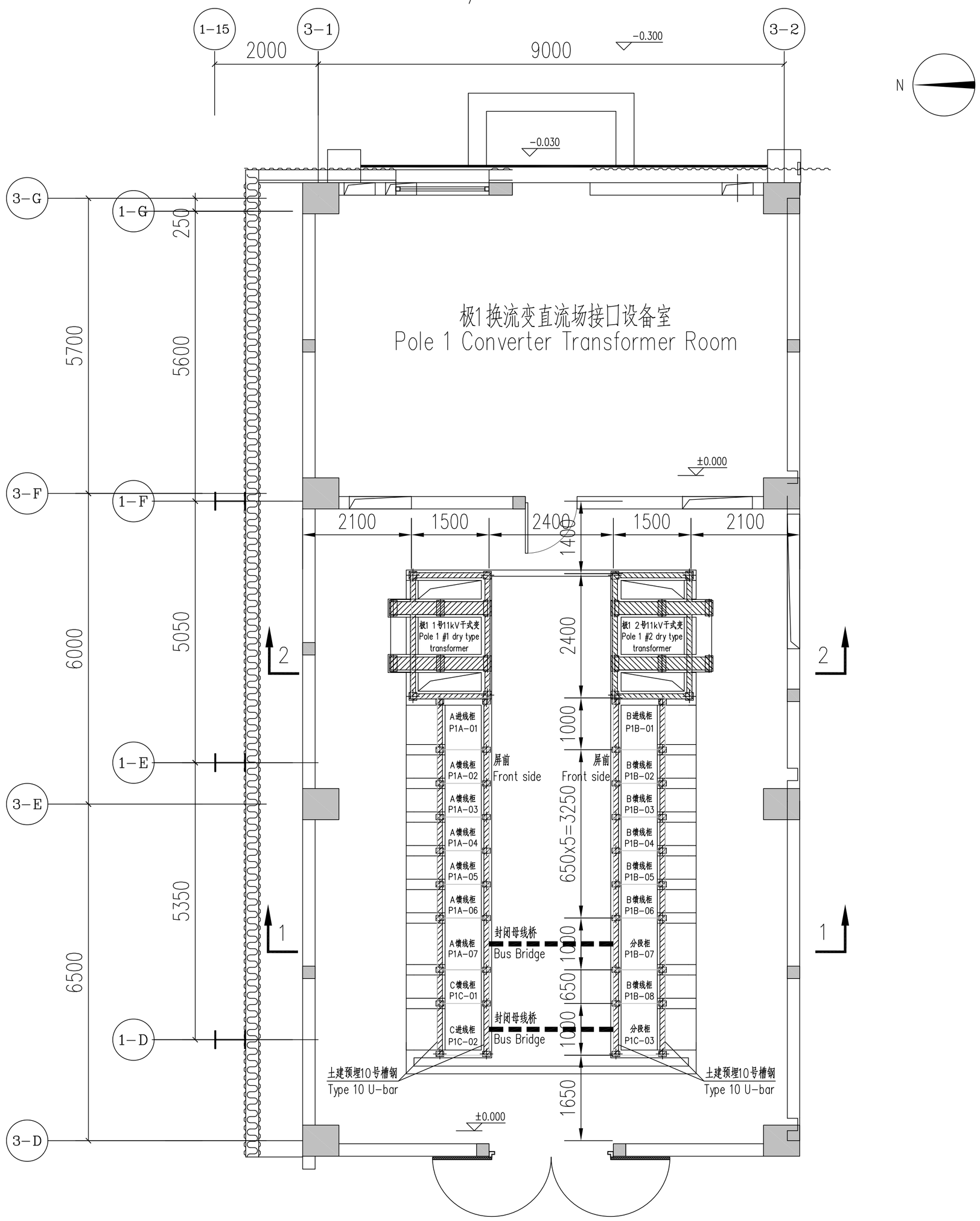
	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	凌建辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益午	2019.05.17

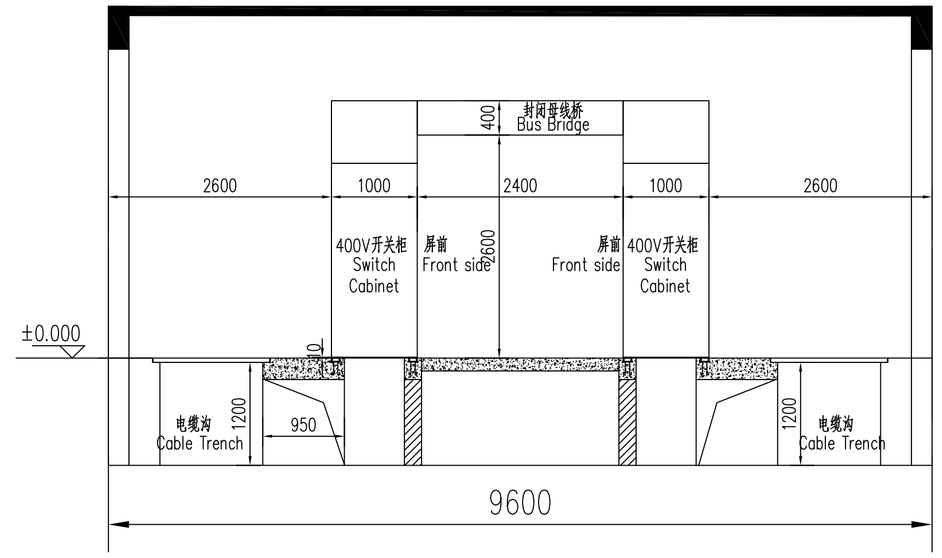
极1 400V站用电系统P1B及P1C段配置图
Configuration drawing of P1B and P1C sections of 400V Station Power System

比例 Scale | 图号 DWG NO. 30-B003701S-D0202-005 | 版本 REV. A

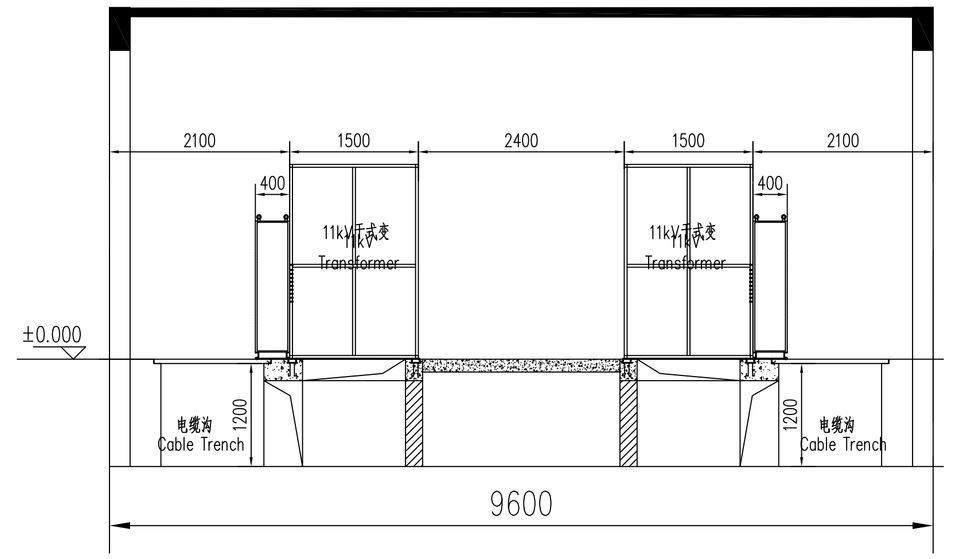
594X841

极1 380/220V交流阀组配电室
Pole 1 380/220V AC transformer room





1-1



2-2

说明：1. 400V开关柜与屏柜底部预埋槽钢螺栓连接。
2. 封闭母线桥柜顶走线，馈线电缆下出线。
3. 400V开关柜PE线和N线在P1A-01、P1B-01、P1C-02屏处采用TJ-150铜绞线与主地网连接，三处接至主地网不同干线，接地材料开列在本图。
4. 11kV干式变采用TJ-150铜绞线与基础下方预留接地引上（规格：50x6镀锌扁钢）连接。

Remarks: 1.The cabinet should be connected to U-bar with bolts.
2. Bus bridge are arranged on top of cabinet, cables are extracted from the bottom of cabinet.
3. The PE and N line of P1A-01、P1B-01、P1C-02 cabinet should be connected to the grounding grid..
4. 50x6 steel are reserved at foundation of transformer for transformer grounding.

材料表
Material table

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	备注 Comments
1	铜绞线 copper stranded conductor	150mm²	米 m	120	
2	镀锡铜鼻子 copper cable lug	DTM 150	套 set	6	
3	放热焊点 welding point	TJ150与TJ120之间 for welding of TJ150 and TJ120	套 set	6	

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

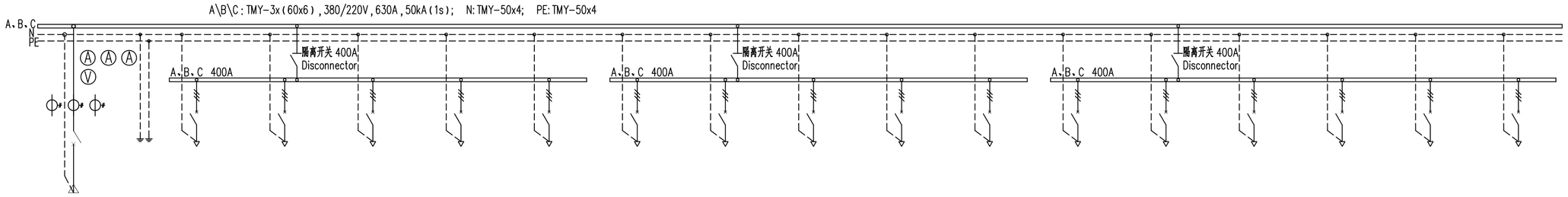
M L H V D C - L C S - E C I - D D - E C - 0 2 0 2 - 0 0 6 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	李建辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益竹	2019.05.17

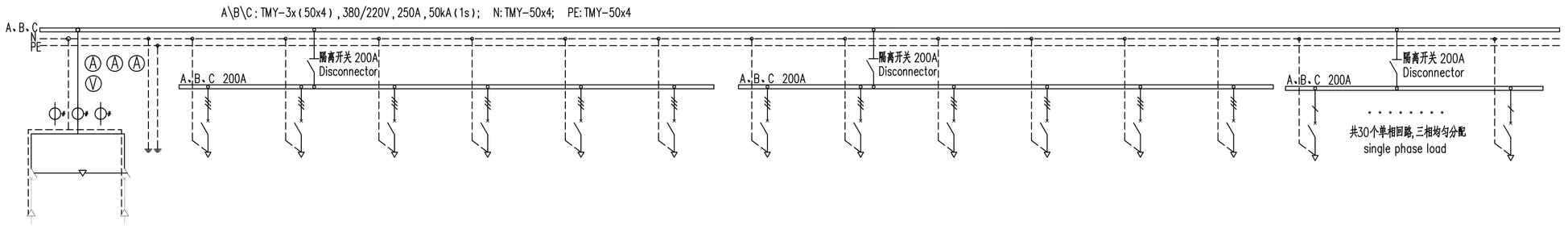
极1 400V站用电室布置图
Layout Drawing of Pole 1 400V Station Service House

比例 Scale:
图号 DWG NO.: 30-B003701S-D0202-006
版本 REV.: A

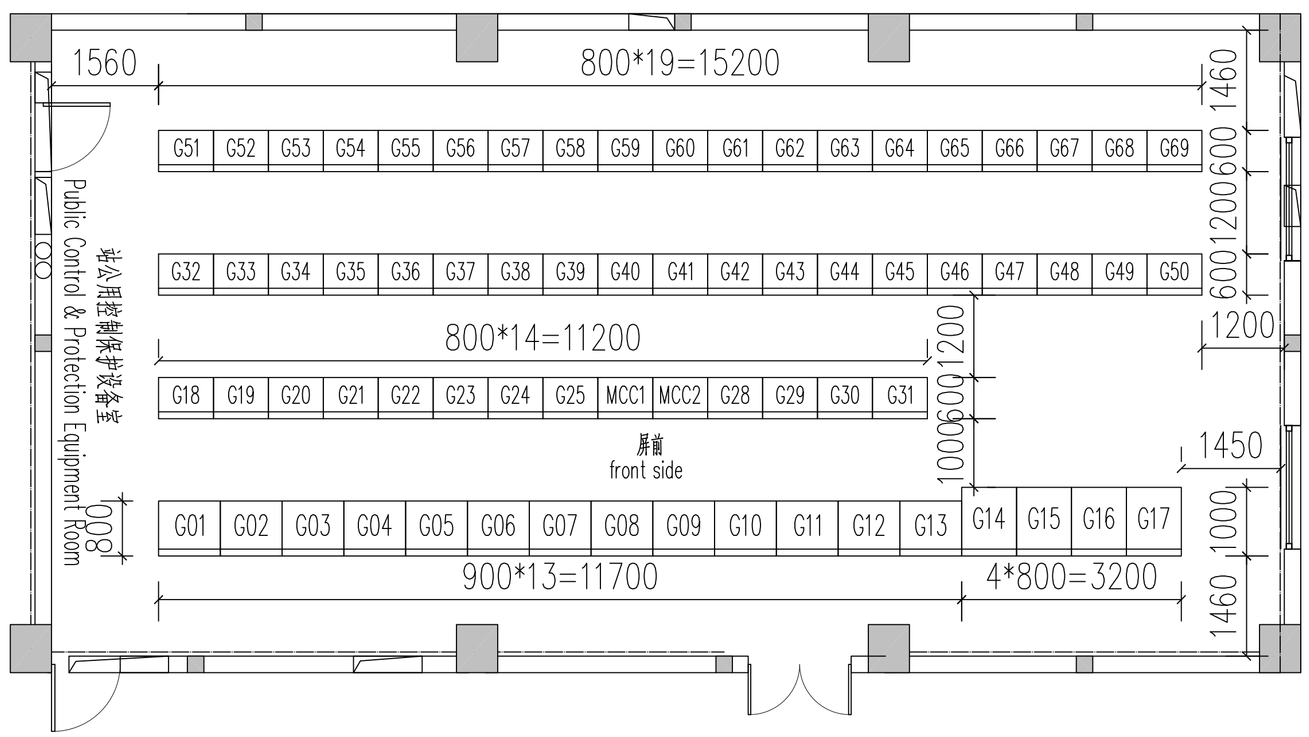
594X841



配电屏编号	主控楼一次MCC1																	
屏宽(mm)	800																	
回路名称	进线	主控楼一层 配电箱GX1 Distribution Box GX1	主控楼一层 配电箱GX2 Distribution Box GX2	主控楼二层 配电箱GX3 Distribution Box GX3	备用 for spare	备用 for spare	主控楼二层 配电箱GX4 Distribution Box GX4	主控楼三层 配电箱GX5 Distribution Box GX5	主控楼楼1 平台车配电箱GX6 Distribution Box GX6	备用 for spare	备用 for spare	主控楼楼2 平台车配电箱GX7 Distribution Box GX7	主控楼楼1 闸冷阀 起重机配电箱GX8 Distribution Box GX8	主控楼楼2 闸冷阀 起重机配电箱GX9 Distribution Box GX9	备用 for spare	备用 for spare	备用 for spare
回路额定容量(kW)																		
回路额定工作电流(A)	400	60	60	60			60	70	80			80	30	30				
断路器型号	T5S630 R630	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R63	XT2S160 R100	XT2S160 R63	XT2S160 R63	XT2S160 R100	XT2S160 R63	XT2S160 R63
断路器额定电流 Iu(A)	630	160	160	160	160	160	160	160	160	160	160	160	160	160	160	160	160
保护电器 脱扣器型号	PR221DS-LSI	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I
保护电器 脱扣器额定电流 In(A)	630	100	100	100			100	100	100			100	63	63			
保护电器 建议长延时脱扣门限值 I1/脱扣曲线时间档位(A/s)	504/3	80/3	80/3	80/3			80/3	90/3	100/3			100/3	50.4/3	50.4/3			
保护电器 建议短延时脱扣门限值 I2/脱扣曲线时间档位(A/s)	3150/0.1	300/0.1	300/0.1	300/0.1			300/0.1	300/0.1	300/0.1			300/0.1	252/0.1	252/0.1			
保护电器 建议瞬动整定电流 I3(A)		关闭 Closed	关闭 Closed	关闭 Closed			关闭 Closed	关闭 Closed	关闭 Closed			关闭 Closed	关闭 Closed	关闭 Closed			
保护电器 建议接地保护门限值 I4/脱扣曲线时间档位(A/s)																	
电流互感器变比	630/1	100/1	100/1	100/1			100/1	100/1	100/1			100/1	75/1	75/1			
相数																	
电缆型号及芯数x截面(mm^2)		ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -3x25+1x16			ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -3x35+1x16	ZRB-YJV22-0.6/1.0 -3x50+1x25			ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x16+1x10			
电缆载流量(空气中50℃)		83.2	83.2	83.2			83.2	100	121.6			121.6	63.2	63.2			
电缆编号(回路编号)	来自站用电屏P1A-03-03	ZKL1-01-01	ZKL1-01-02	ZKL1-01-03	ZKL1-01-04	ZKL1-01-05	ZKL1-01-06	ZKL1-01-07	ZKL1-01-08	ZKL1-01-09	ZKL1-01-10	ZKL1-01-11	ZKL1-01-12	ZKL1-01-13	ZKL1-01-14	ZKL1-01-15	ZKL1-01-16
电缆长度(米)		61	84	25			35	97	62			84	90	95			



配电屏编号	主控楼二次MCC2																	
屏宽(mm)	800																	
回路名称	进线		移动式试验电源装置1 Testing power pannel of Control Building 1	备用 for spare	主控楼事故照明屏 Emergency Lighting pannel of Control Building	备用 for spare	备用 for spare	备用 for spare	移动式试验电源装置2 Testing power pannel of Control Building 2	备用 for spare	备用 for spare	备用 for spare	备用 for spare	备用 for spare	二次单相回路1 single phase load 1	二次单相回路n single phase load n	二次单相回路30 single phase load 30
回路额定容量(kW)			15		15(单相)				15								
回路额定工作电流(A)	~120		30		75				30								
断路器型号	ATS-CB010/XT4S250 R250		XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R63	XT2S160 R63	XT2S160 R63	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R63	S201-C32	S201-C32	S201-C32
断路器额定电流 Iu(A)	250	250	160	160	160	160	160	160	160	160	160	160	160	160			
保护电器 脱扣器型号	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I			
保护电器 脱扣器额定电流 In(A)	250	250			100												
保护电器 建议长延时脱扣门限值 I1/脱扣曲线时间档位(A/s)	180/3	180/3			90/3												
保护电器 建议短延时脱扣门限值 I2/脱扣曲线时间档位(A/s)	1000/0.1	1000/0.1			300/0.1												
保护电器 建议瞬动整定电流 I3(A)	关闭 Closed	关闭 Closed			关闭 Closed												
保护电器 建议接地保护门限值 I4/脱扣曲线时间档位(A/s)																	
电流互感器变比	250/1	250/1			100/1												
相数															共30个单相回路,三相均匀分配 30 single phase load		
电缆型号及芯数x截面(mm^2)					ZRB-YJV22-0.6/1.0 -2x35												
电缆载流量(空气中50℃)					110.4												
电缆编号(回路编号)	来自站用电屏P1A-03-01	来自站用电屏P1B-03-01	ZKL2-01-01	ZKL2-01-02	ZKL2-01-03	ZKL2-01-04	ZKL2-01-05	ZKL2-01-06	ZKL2-01-07	ZKL2-01-08	ZKL2-01-09	ZKL2-01-10	ZKL2-01-11	ZKL2-01-12			
电缆长度(米)					50												



主控楼MCC1及MCC2交流分屏布置图

Layout drawing of Control Building

说明:1. 站前区办公楼配电室交流分屏位置如图所示。
2. 交流分屏本期一次配置完成,采用固定式。
3. 交流分屏馈线中的二次单相回路电缆由二次专业设计并开列,见二次专业相关图纸。
4. PR221长延时脱扣曲线时间档位所对应的电流值为@6I1,短延时脱扣曲线时间档位所对应的电流值为@8In;
5. LS/I长延时脱扣曲线时间档位所对应的电流值为@3I1,短延时脱扣曲线时间档位所对应的电流值为@10In。
6. 交流分屏MCC1、MCC2与基础预埋槽钢焊接安装,N线、PE线分别通过TJ-150铜绞线与户内环房接地干线放热焊接。
7. 电流载流量根据重庆泰山电缆有限公司提供的资料确定(按空气中50°环温考虑,并考虑0.8的敷设系数)。
8. 图中开关保护整定值仅供参考,现场可根据实际情况调整。

Descriptions:1. The location of MCC1 and MCC2 are shown as Layout drawing of Distribution Room.
2. Circuit breakers in MCC1 and MCC2 are fixed type.
3. The single phase circuits are designed by the secondary electrical major.
4. The time of long delay overload tripping curve of PR221 is @6I1.
The time of short delay overload tripping curve of PR221 is @8In.
5. The time of long delay overload tripping curve of LS/I is @3I1.
The time of short delay overload tripping curve of LS/I is @10In.
6. Switch cabinet MCC1 and MCC2 should be welded to U-bar of the foundation when installing.
7. Current-carrying capacity of cable is according to cable manufacture.
8. The settings of tripping device are for reference.

材料表
Material table

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	备注 Comments
1	铜绞线 copper stranded conductor	150mm²	米 m	60	
2	镀锡铜鼻子 copper cable lug	DTM 150	套 set	6	
3	放热焊点 welding point	TJ150与TJ120之间 for welding of TJ150 and TJ120	套 set	6	

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

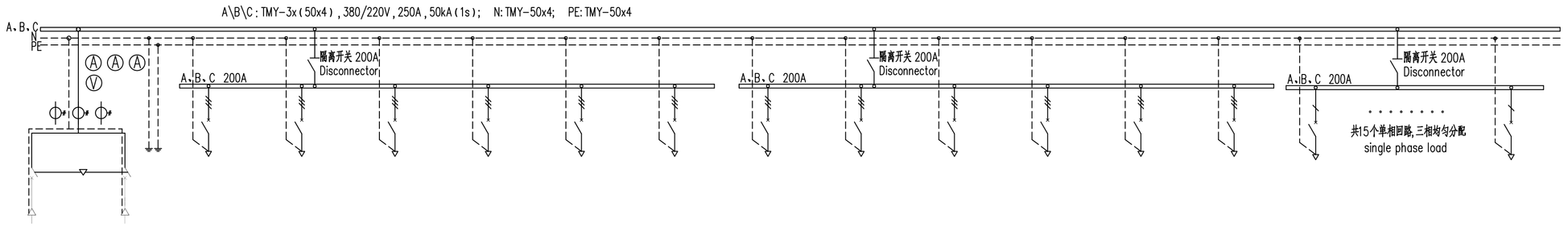
M L H V D C - L C S - E C I - D D - E C - 0 2 0 2 - 0 0 7 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	凌艳辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益竿	2019.05.17

主控楼MCC1及MCC2交流分屏配置及布置图
Configuration and Layout Drawing MCC1 and MCC2 of Control Building

比例 Scale | 图号 DWG NO. 30-B003701S-D0202-007 | 版本 REV. A

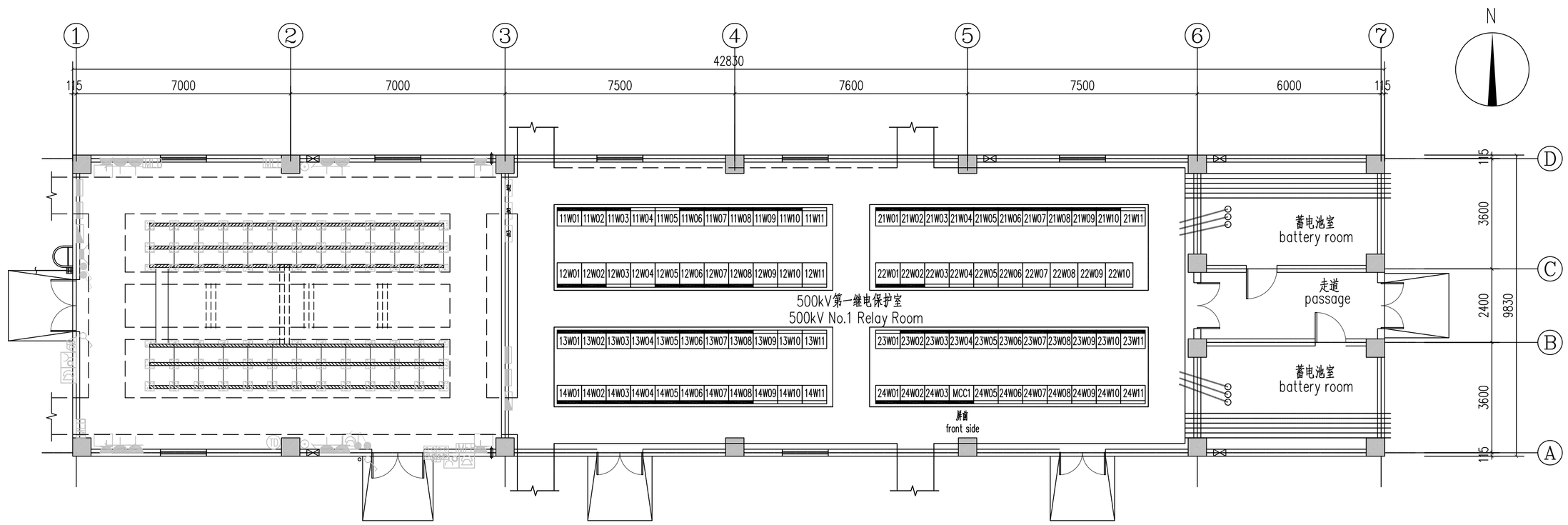
594X841



配电屏编号			#1继电器室MCC1														
屏宽(mm)			800														
回路名称	进线		#1继电器室配电箱J1AL1 Distribution Box J1AL1	#1继电器室配电箱J1AP2 Distribution Box J1AP2	#1继电器室配电箱J1AL2 Distribution Box J1AL2	备用 for spare	备用 for spare	500kV开关场户外照明配电箱4 Light Power Box HWZMX4	#1继电器室试验电源屏1 No.1 testing power pannel	#1继电器室试验电源屏2 No.2 testing power pannel	#1继电器室事故照明屏 Emergency Lighting pannel	备用 for spare	备用 for spare	备用 for spare	二次单相回路1 single phase load 1	二次单相回路n single phase load n	二次单相回路15 single phase load 15
回路额定容量(kW)			22	15	14			16.76	15	15	5kVA(单相)						
回路额定工作电流(A)	~180		34	23	22			38.2	30	30	25						
断路器型号	ATS-CB010/T3S250 R250		XT2S160 R63	XT2S160 R63	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R63	XT2S160 R63	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R63	S201-C32	S201-C32	S201-C32
断路器额定电流 Iu(A)	250	250	160	160	160	160	160	160	160	160	160	160	160	160			
保护电器 脱扣器型号	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I			
保护电器 脱扣器额定电流 In(A)	250	250	63	63	63			100	63	63	63						
保护电器 建议长延时脱扣门限值 I1/脱扣曲线时间档位(A/s)	200/3	200/3	44.1/3	31.5/3	31.5/3			60/3	37.8/3	37.8/3	37.8/3						
保护电器 建议短延时脱扣门限值 I2/脱扣曲线时间档位(A/s)	500/0.1	500/0.1	126/0.1	126/0.1	126/0.1			200/0.1	126/0.1	126/0.1	126/0.1						
保护电器 建议瞬动整定电流 I3(A)	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed			关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed						
保护电器 建议接地保护门限值 I4/脱扣曲线时间档位(A/s)																	
电流互感器变比	250/1	250/1	75/1	75/1	75/1			100/1	75/1	75/1	75/1						
相数															共15个单相回路,三相均匀分配 15 single phase load		
电缆型号及芯数x截面(mm^2)			ZRB-YJV22-0.6/1.0 -4x10	ZRB-YJV22-0.6/1.0 -4x10	ZRB-YJV22-0.6/1.0 -4x10			ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -4x10	ZRB-YJV22-0.6/1.0 -4x10	ZRB-YJV22-0.6/1.0 -2x10(脱A相)						
电缆载流量(空气中50℃)			47.2	47.2	47.2			83.2	47.2	47.2	53.6						
电缆编号(回路编号)	来自站用电屏P1A-06-13	来自站用电屏P1B-06-12	J1-01-01	J1-01-02	J1-01-03	J1-01-04	J1-01-05	J1-01-06	J1-01-07	J1-01-08	J1-01-09	J1-01-10	J1-01-11	J1-01-12			
电缆长度(米)			25	48	46			69	16	16	11						

说明:1. #1继电器室MCC1交流分屏位置如图所示。
2. 交流分屏本期一次配置完成,采用固定式。
3. 交流分屏馈线中的二次单相回路电缆由二次专业设计并开列,见二次专业相关图纸。
4. LS/I长延时脱扣曲线时间档位所对应的电流值为@3I1,短延时脱扣曲线时间档位所对应的电流值为@10In。
5. 交流分屏MCC1与基础预埋槽钢焊接安装,N线、PE线分别通过TJ-150铜绞线与户外主地网放热焊接。
6. 电流载流量根据重庆泰山电缆有限公司提供的资料确定(按空气中50°环温考虑,并考虑0.8的敷设系数)。
7. 图中开关保护整定值仅供参考,现场可根据实际情况调整。

Descriptions:1. The location of MCC1 is shown as Layout drawing of 500kV No.1 Relay Room.
2. Circuit breakers in MCC1 are fixed type.
3. The single phase circuits are designed by the secondary electrical major .
4. The time of long delay overload tripping curve of LS/I is @3I1.
The time of short delay overload tripping curve of LS/I is @10In.
5. Switch cabinet MCC1 should be welded to U-bar of the foundation when installing.
6. Current-carrying capacity of cable is according to cable manufacture.
7. The settings of tripping device are for reference.



#1继电器室MCC1交流分屏布置图

Layout drawing of 500kV No.1 relay room

材料表
Material table

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	备注 Comments
1	铜绞线 copper stranded conductor	150mm²	米 m	30	
2	镀锡铜鼻子 copper cable lug	DTM 150	套 set	2	
3	放热焊点 welding point	TJ150与TJ120之间 for welding of TJ150 and TJ120	套 set	2	

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 2 0 2 - 0 0 8 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	李建辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益午	2019.05.17

#1继电器室MCC1交流分屏配置及布置图
Configuration and Layout Drawing MCC1 of 500kV No.1 Relay Room

比例 Scale | 图号 DWG NO. 30-B003701S-D0202-008 | 版本 REV. A

594X841